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Bihar Engineering University, Patna

End Semester Examination - 2023

Time: 03 Hours

Code: 100403 ***Subject: Digital Electronics*** ***Full Marks: 70***

Instructions:-

(i) The marks are indicated in the right-hand margin.
(ii) There are NINE questions in this paper.
(iii) Attempt FIVE questions in all.
(iv) Question No. 1 is compulsory.

Q.1 Answer any seven Question only: **[14]**

(a) Draw Truth table of JK flip flop.

(b) Convert the Decimal number (108.025) to their binary equivalent.

(c) Draw the truth table and logic circuit of Half-adder.

(d) Write the difference between combinational & Sequential circuit.

(e) Draw timing diagram of SR flip-flop.

(f) Find 2's complement of 1011011.

(g) Number of 2:1 mux requires designing 256:1 mux is

(h) Add hexadecimal number 2ABC & 98F2.

(i) Discuss Universal gates.

(j) What is the use of K-map?

Q.2 Solve this question : **[14]**

Implement the following function using only NAND gate F(A,*B*,C)=Σm(0,1,2,3,7).

Q.3 Solve both questions : **[7+7=14]**

(a) Realize XNOR logic function using NAND gate only.

(b) Simplify *Y=ABC+AB*overlineC+Aoverline*B*C

Q.4 Solve this question : **[14]**

A logic circuit has four inputs A,B,C,D and output Y. Y=1 when A & B are both 1, subjected to the condition that C and D are both low or both high. Design the logic circuit.

Q.5 Solve this question : **[14]**

Explain Master-slave flip flop. What are race around condition?. How it can be circumvented with the help of Master-slave flip flop.

Q.6 Solve both questions : **[7+7=14]**

(a) Design 8 to 3 line Encoder circuit.

(b) Implement NAND gate using TTL logic family.

Q.7 Solve this question : **[14]**

Summarize the design procedure for a synchronous sequential circuit.

Q.8 Solve both questions : **[7+7=14]**

(a) Explain the working of R-2R Ladder DAC.

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(b) Design 3-bit binary counter using T flip-flop.

Q.9 Write short notes on the following: **[14]**

(a) (a) Binary Parallel Adder

(b) (b) Digital IC logic families

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